



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SANKAR AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

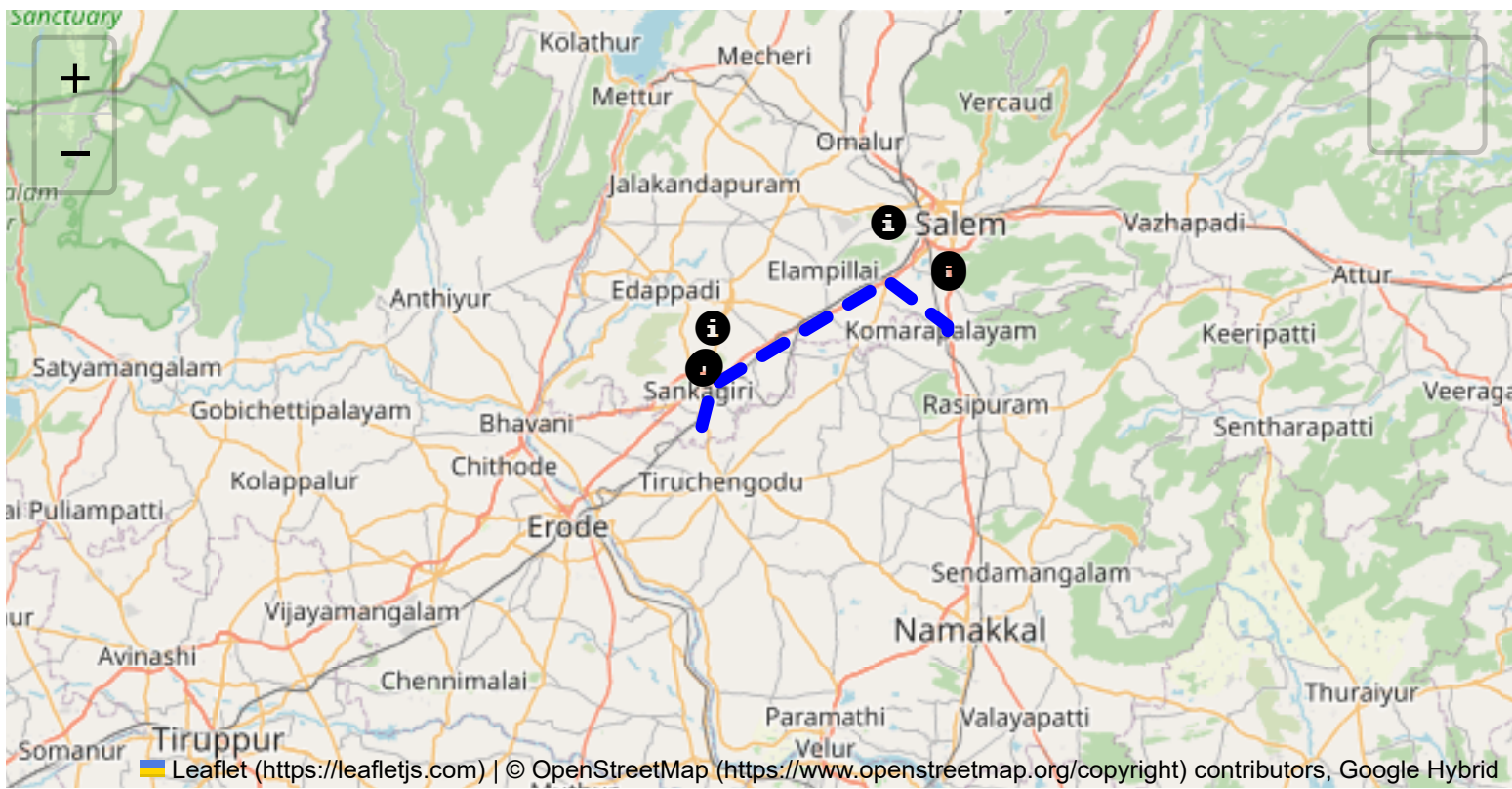
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

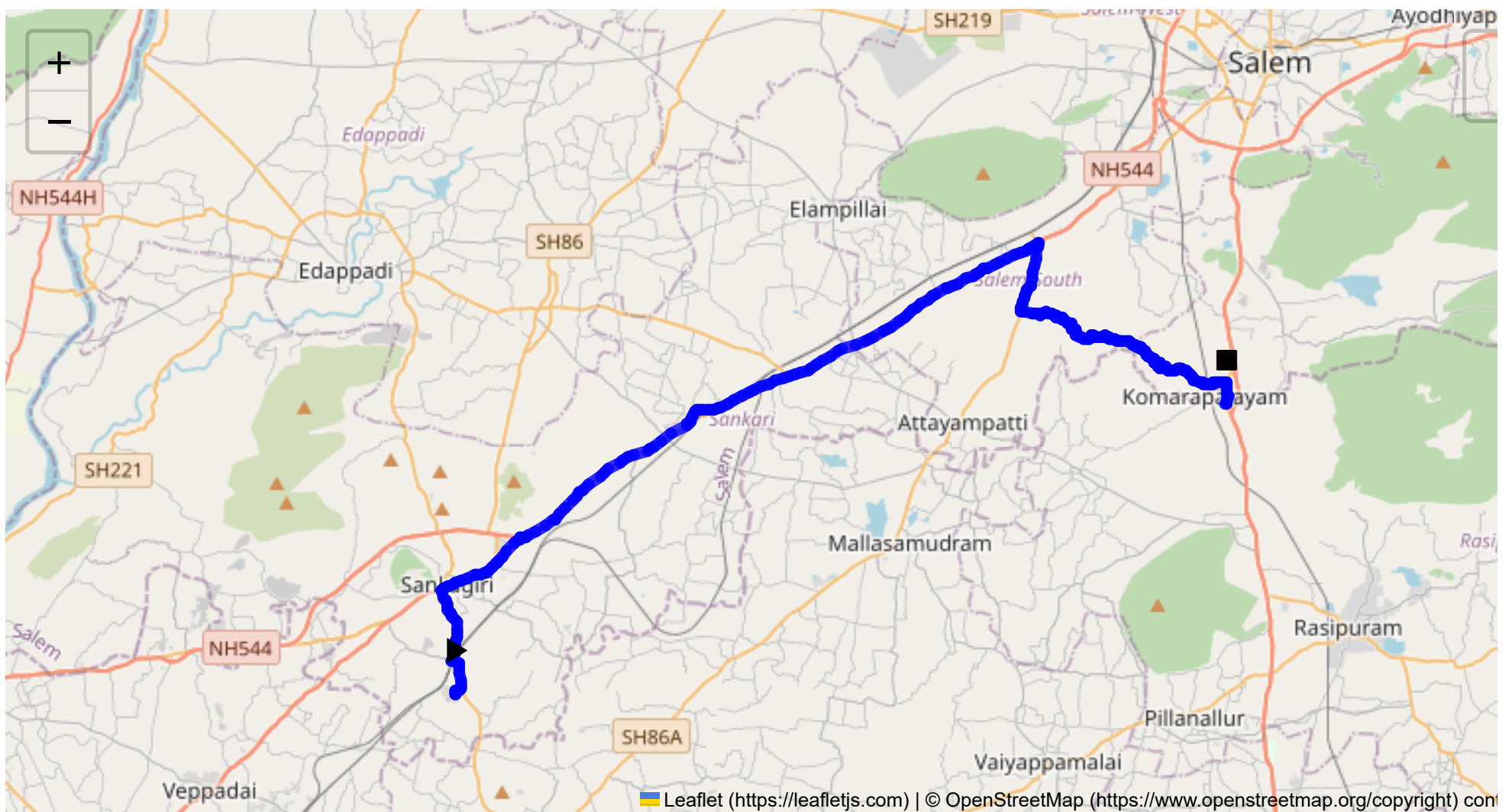
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.76 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.53807, 78.142586)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route begins at Sangagiri in Tamil Nadu, leading through Kangeyampalayam, Akkamapettai-Sendrayagovandanur Rd, and Ariyanur, to reach Mallur. This route largely follows state highways and local roads through rural and semi-urban areas, covering a distance of approximately 44.76 kilometers.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu generally experiences a tropical climate. The region can be subject to heavy rains during the monsoon season (October to December) which may lead to waterlogging and reduced visibility. Heat can be intense in the summer months, potentially affecting vehicle performance.

3. Analysis of Traffic Patterns:

1. Congestion is likely around Sangagiri and Ariyanur, especially during market hours and weekends.
2. Peak congestion hours typically occur during the morning (8-10 AM) and evening (5-7 PM) commutes.
3. Traffic decreases in rural sections between towns.

4. Assessment of Road Quality and Infrastructure:

1. The primary roads are generally well-maintained. However, some secondary roads may have potholes and uneven surfaces.
2. Rural roads might be narrow, which can pose challenges for heavy vehicles.

5. Suggestions for Alternative Routes for Emergencies: In case of roadblocks or emergencies, consider rerouting through the highways NH44 or NH544 for faster detours to neighboring towns or regions.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

1. Tamil Nadu requires special permits for transporting hazardous materials.
2. Designated routes for transporting dangerous goods should be adhered to, and vehicles should carry signs indicating hazardous material.

7. Overview of Historical Incidents:

1. Reports of road mishaps involving heavy vehicles are relatively rare but have occurred, often related to improper handling or mechanical failure.
2. No major chemical spills or significant incidents have been documented recently.

8. Environmental Considerations and Sensitive Areas:

1. The route passes through agricultural zones; contamination from hazardous materials could impact crops and water sources.
2. Efforts must be made to minimize noise and air pollution, especially near village settlements.

9. Analysis of Communication Coverage:

1. Mobile and GPS coverage is generally good along the main route.
2. Some rural pockets may experience weak signal, impacting communication.

10. Estimated Emergency Response Times:

1. Urban areas (Sangagiri, Ariyanur, Mallur) can expect an emergency response time of approximately 15-30 minutes.
2. Remote sections could see responses delayed to 45 minutes or more.

12. Overall Summary of Risk Assessment: The route from Sangagiri to Mallur offers a reasonably safe path for truck drivers, with standard risks associated with rural and semi-urban transit. Major risks include road quality fluctuations, congestion during peak hours, and weather impacts. With proper precautions, adherence to local regulations, and awareness of emergency protocols, drivers can manage these risks effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44028, 77.87541	15 KM/Hr
3	Turn	Medium	11.44898, 77.87410	30 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	High	11.49148, 77.89601	15 KM/Hr
7	Turn	High	11.49224, 77.89601	15 KM/Hr
8	Blind Spot	Blind Spot	11.59334, 78.07682	10 KM/Hr
9	Turn	Medium	11.59294, 78.07698	30 KM/Hr
10	Blind Spot	Blind Spot	11.56939, 78.07098	10 KM/Hr
11	Turn	Medium	11.56926, 78.07974	30 KM/Hr
12	Turn	High	11.56842, 78.08324	15 KM/Hr
13	Turn	Medium	11.56830, 78.08332	30 KM/Hr
14	Turn	Medium	11.56769, 78.08330	30 KM/Hr
15	Turn	High	11.56757, 78.08338	15 KM/Hr
16	Turn	Medium	11.56751, 78.08513	30 KM/Hr
17	Turn	Medium	11.56735, 78.08530	30 KM/Hr
18	Turn	Medium	11.56702, 78.08540	30 KM/Hr
19	Turn	High	11.56519, 78.08850	15 KM/Hr
20	Turn	High	11.56330, 78.08888	15 KM/Hr
21	Turn	Medium	11.56321, 78.08906	30 KM/Hr
22	Turn	High	11.56325, 78.08996	15 KM/Hr
23	Turn	Medium	11.56133, 78.09013	30 KM/Hr
24	Turn	Medium	11.56016, 78.09452	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
25	Turn	Medium	11.56038, 78.09489	30 KM/Hr
26	Turn	High	11.56115, 78.09531	15 KM/Hr
27	Turn	Medium	11.56063, 78.10155	30 KM/Hr
28	Turn	Medium	11.56035, 78.10183	30 KM/Hr
29	Turn	Medium	11.56015, 78.10333	30 KM/Hr
30	Turn	Medium	11.55949, 78.10566	30 KM/Hr
31	Turn	Medium	11.55914, 78.10890	30 KM/Hr
32	Turn	Medium	11.55896, 78.10903	30 KM/Hr
33	Turn	High	11.55865, 78.10903	15 KM/Hr
34	Turn	Medium	11.55845, 78.10927	30 KM/Hr
35	Turn	Medium	11.55816, 78.11056	30 KM/Hr
36	Turn	High	11.55747, 78.11245	15 KM/Hr
37	Turn	High	11.55676, 78.11262	15 KM/Hr
38	Turn	Medium	11.55618, 78.11434	30 KM/Hr
39	Turn	Medium	11.55460, 78.11486	30 KM/Hr
40	Turn	Medium	11.55442, 78.11508	30 KM/Hr
41	Turn	Medium	11.55349, 78.11730	30 KM/Hr
42	Turn	High	11.55212, 78.11762	15 KM/Hr
43	Blind Spot	Blind Spot	11.55182, 78.11947	10 KM/Hr
44	Turn	Medium	11.55090, 78.11894	30 KM/Hr
45	Turn	Medium	11.55071, 78.11891	30 KM/Hr
46	Turn	High	11.55058, 78.11897	15 KM/Hr
47	Turn	Medium	11.55032, 78.11990	30 KM/Hr
48	Turn	Medium	11.54911, 78.12263	30 KM/Hr
49	Turn	Medium	11.54911, 78.12284	30 KM/Hr
50	Turn	Medium	11.55006, 78.12608	30 KM/Hr
51	Turn	High	11.55025, 78.12650	15 KM/Hr
52	Turn	Medium	11.54952, 78.12857	30 KM/Hr
53	Turn	High	11.54852, 78.12910	15 KM/Hr
54	Turn	High	11.54818, 78.12992	15 KM/Hr
55	Turn	Medium	11.54768, 78.13016	30 KM/Hr
56	Turn	Medium	11.54689, 78.13130	30 KM/Hr
57	Turn	Medium	11.54604, 78.13152	30 KM/Hr
58	Turn	Medium	11.54520, 78.13411	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
59	Turn	High	11.54498, 78.13426	15 KM/Hr
60	Turn	Medium	11.54499, 78.13472	30 KM/Hr
61	Turn	Medium	11.54459, 78.13696	30 KM/Hr
62	Turn	High	11.54457, 78.13802	15 KM/Hr
63	Turn	Medium	11.54518, 78.13843	30 KM/Hr
64	Blind Spot	Blind Spot	11.54499, 78.14302	10 KM/Hr
65	Blind Spot	Blind Spot	11.53809, 78.14305	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
2	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
3	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
4	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	police	Mallur Police Station	11.5452987, 78.1428828	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44028, 77.87541



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47407, 77.86839



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49148, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.59334, 78.07682



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59294, 78.07698



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.56939, 78.07098



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56926, 78.07974



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.56842, 78.08324



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56830, 78.08332



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56769, 78.08330



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.56757, 78.08338



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56751, 78.08513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56735, 78.08530



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56702, 78.08540



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56519, 78.08850



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.56330, 78.08888



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56321, 78.08906



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.56325, 78.08996



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56133, 78.09013



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56016, 78.09452



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56038, 78.09489



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.56115, 78.09531



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56063, 78.10155



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56035, 78.10183



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56015, 78.10333



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55949, 78.10566



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55914, 78.10890



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55896, 78.10903



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.55865, 78.10903



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55845, 78.10927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55816, 78.11056



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55747, 78.11245



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55676, 78.11262



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55618, 78.11434



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55460, 78.11486



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55442, 78.11508



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55349, 78.11730



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55212, 78.11762



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.55182, 78.11947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55090, 78.11894



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55071, 78.11891



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55058, 78.11897



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55032, 78.11990



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54911, 78.12263



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54911, 78.12284



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55006, 78.12608



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55025, 78.12650



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54952, 78.12857



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.54852, 78.12910



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.54818, 78.12992



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54768, 78.13016



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54689, 78.13130



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54604, 78.13152



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54520, 78.13411



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.54498, 78.13426



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54499, 78.13472



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54459, 78.13696



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.54457, 78.13802



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54518, 78.13843



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.54499, 78.14302



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.53809, 78.14305
